HOME-LOAN BOOK · RISK, FRAUD CONTROL AND COLLECTIONS

Early Warning, Smart Collections

Risk and collections teams moved from reacting to overdue buckets to acting on early signals: a score for every live loan, link analysis that exposes fraud rings, capacity-aware collections and outreach checked for fair practice before it is sent.

~~130~~ → **101** per 10,000 rolling from 1–30 to 31–60 days past due

45 days
Earlier warning of stress
before the first missed EMI

22 % fewer
Accounts rolling 1–30 → 31–60 DPD
▼ more customers cured early

2.5×
Linked suspicious applications caught before disbursal
network checks, not file-by-file

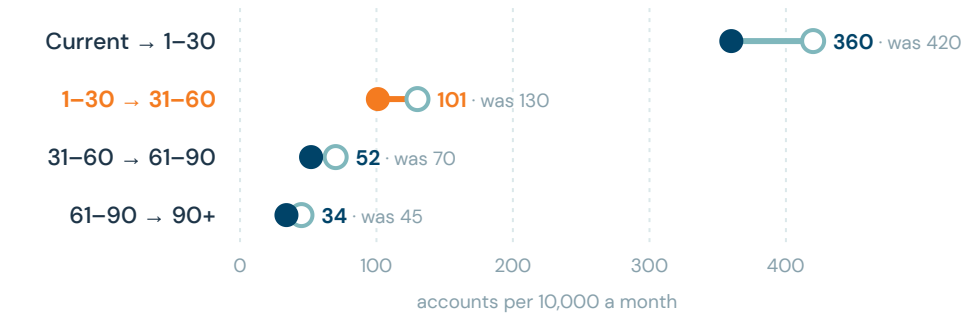
18 % lower
Collection cost per ₹100 recovered
▼ right channel for each customer

Rings: one segment per monthly signal in the score (seven); roll rate after (101) against before (130, inside); fraud catch at 2.5× with the earlier level inside; cost per ₹100 after against before (inside). Figures indicative; validated metrics available on request.

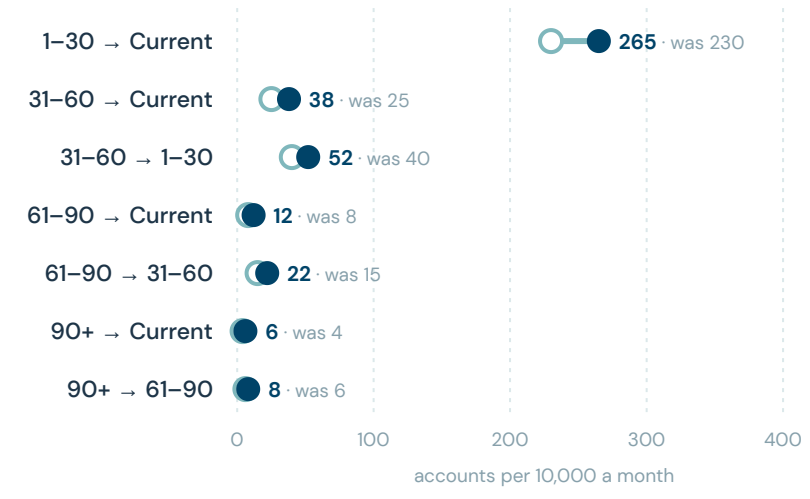
01 Fewer accounts roll forward, more cure early

Monthly movement between days-past-due buckets · before (open) → after (filled)

ROLLED FORWARD (WORSE)



CURED OR MOVED BACK



~~665~~ **547** ALL FORWARD ROLLS
per 10,000 a month; the 1–30 → 31–60 step, where the best moment used to pass, falls 22%.

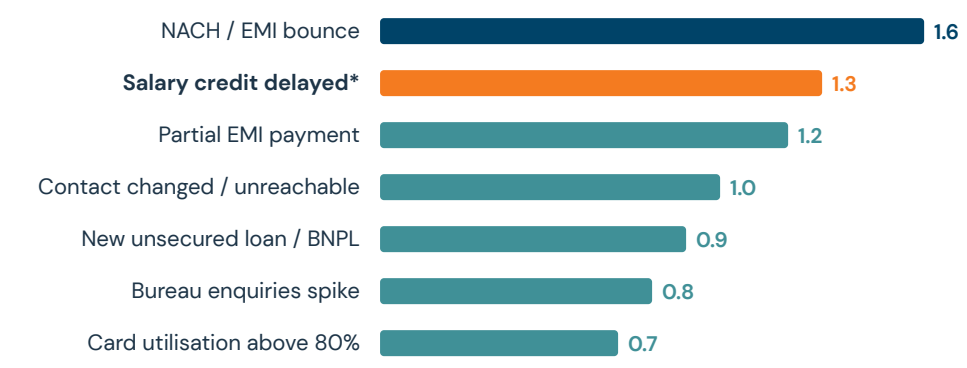
~~328~~ **403** ALL CURES AND STEP-BACKS
per 10,000 a month; every backward flow grows once stress is acted on before the bucket changes.

02 A score that reads stress before the EMI is missed

Early-warning radar · seven monthly signals, every score shows its reasons

WHAT THE SCORE READS

Repayment, bureau, bank-statement and contact signals only; no protected attributes. Weight per signal:



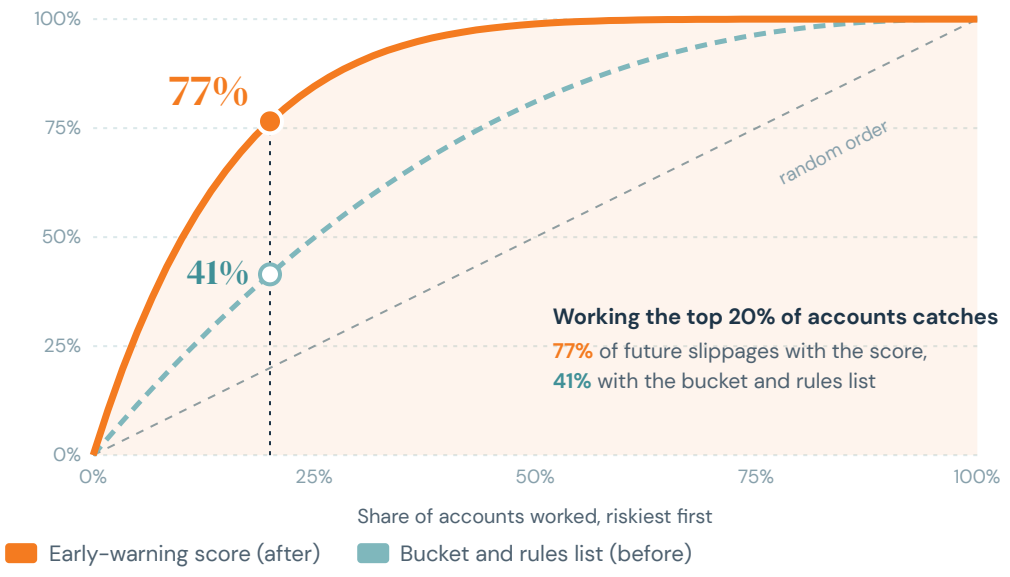
RISK BANDS

Low · 0%+	Watch · 8%+	High · 20%+	Critical · 45%+

Weights as in the radar's score · *business credits down 30%+ for the self-employed

HOW WELL THE LIST FINDS FUTURE SLIPPAGES

Cumulative gains: share of slippages caught by working the riskiest accounts first.



03 Four pain points, four fixes

Before → after

BEFORE
Late detection
Stress showed up as a missed EMI. Salary delays, bounces and new borrowing were not tracked together.

EARLY-WARNING SCORE

AFTER
Weeks earlier
Seven monthly signals score every live loan; high-risk accounts get a pre-due call.

BEFORE
Fraud hidden in the links
Each application looked clean alone. Shared phones, devices, valuers and employers only showed across files.

LINK GRAPH

AFTER
Rings before disbursal
Links across applications; the fraud control unit reviews every flagged ring. Nothing is rejected automatically.

BEFORE
One-size collections
Lists worked by bucket and amount. Self-curers got calls; those needing a plan got pressure.

PROPENSITY + CAPACITY

AFTER
Right treatment
Allocated by ability, willingness and team capacity; hardship cases routed to counsellors.

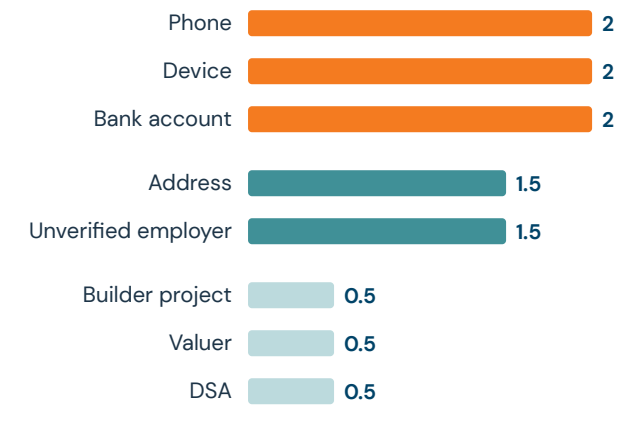
BEFORE
Conduct risk
Fair-practice checks on timing, tone and privacy were manual and sample-based, after the fact.

GENAI + CHECKER

AFTER
Every message checked
Personal outreach in the customer's language, checked before it is sent.

04 Which links count

Link weight between two applications · fraud ring detector



Strong · **medium** · **mostly benign** link weight
3+ linked applications = a ring, sent to verification

05 Treatment matrix

Overdue and high-risk accounts by ability and willingness to pay

WILLINGNESS →	ABILITY TO PAY →
Wants to pay, struggling → hardship plan with a counsellor ₹900 per case · 5 per cycle	**Can pay, will pay** → digital nudge ₹5 per nudge · no limit
Struggling and unreachable → field visit ₹600 per visit · 6 per cycle	**Can pay, avoiding** → firm tele-calling ₹60 per call · 15 per cycle

Costs as modelled per contact; capacity per tele-caller, field agent or counsellor each cycle. Same team, propensity-led allocation lowers cost per ₹100 recovered.

06 How HMC ran it

2 weeks — Discover
Back-tested past slippages; signals, fraud cases and channel costs mapped.

3 weeks — Design
Explainable score, link rules and treatment matrix, signed off by risk and compliance.

6 weeks — Build
Risk radar, link graph, allocation planner and message checker; daily dashboards.

Pilot zones — Adopt
People decide, the tools prepare: pre-due calls, ring reviews, agent training.

∞ ongoing — Scale
All zones; monthly drift and fairness checks, new link types added.

07 Nine checks before any message goes out

Collections copilot · fair-practice gate on every GenAI draft

DRAFTED FOR EACH CUSTOMER

Channels	**WhatsApp · SMS · call script · email**
Languages	**English, Hindi + six more**
Tone	**Gentle · hardship · firm**
Replies read for	**promise, hardship, dispute**

CONTACT WINDOW

00:00 · 08:00 · 19:00 · 24:00

1. **Lender identified** — Name the lender up front.
2. **Amount and due date** — State what is due and when.
3. **Options to pay** — A payment link, plan or EMI-date change.
4. **Help and grievance** — Customer care and grievance contacts.
5. **No threats** — No police, arrest or false legal claims.
6. **No third-party disclosure** — Never employers, family or neighbours.
7. **Respectful language** — No labels such as "defaulter".
8. **Contact time 08:00–19:00** — Nothing outside the window.
9. **Agent introduces self** — Call scripts open with name and lender.

All pass → ready to send · **Any fail → blocked until fixed** · Orange: checks once left to sample audits.

What any lender can borrow

01 — Signals beat buckets
The month before a missed EMI usually tells the story. Score it, explain it and act before the bucket changes.

02 — Fraud hides in the links
A clean file can sit inside a dirty network. Check what applications share, and know which links are harmless.

03 — Right treatment, within the rules
Match the channel to the customer, help those in hardship, and check every message for fair practice before it goes out.

Built with **Python** · **Gradient boosting** with reason codes · **Graph analytics** · **Optimisation** · **Large language model** · **Power BI** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.